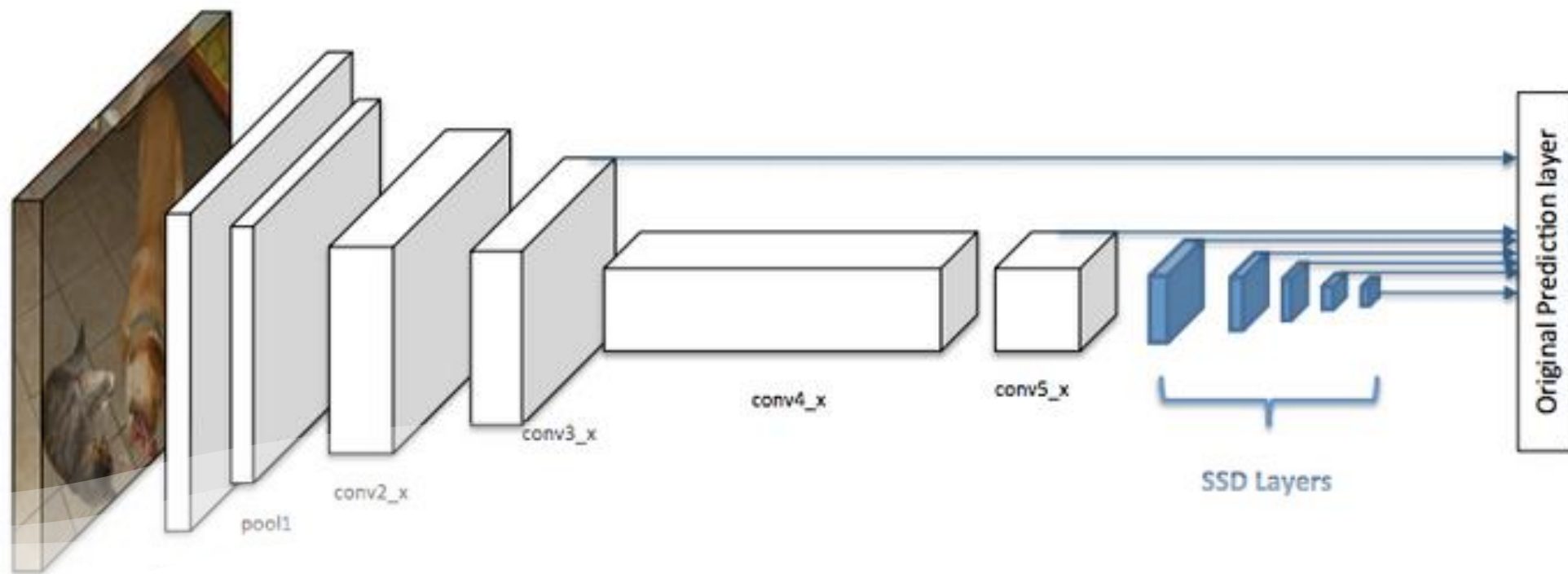




SSD – backbone (VGG) and SSD head

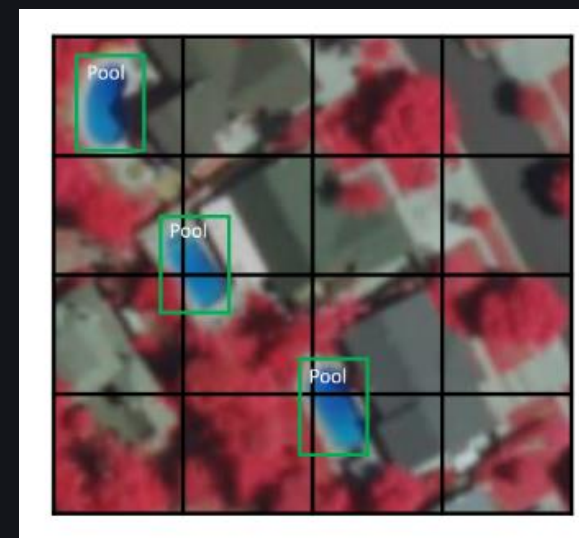
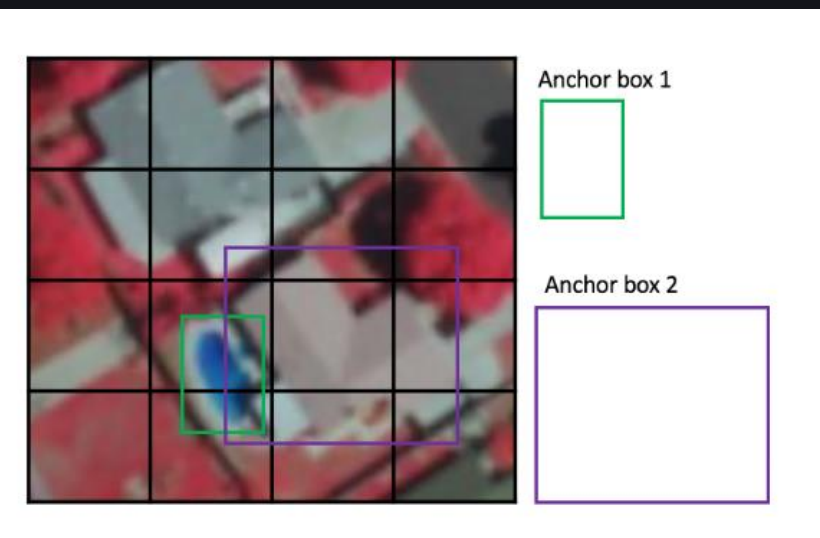
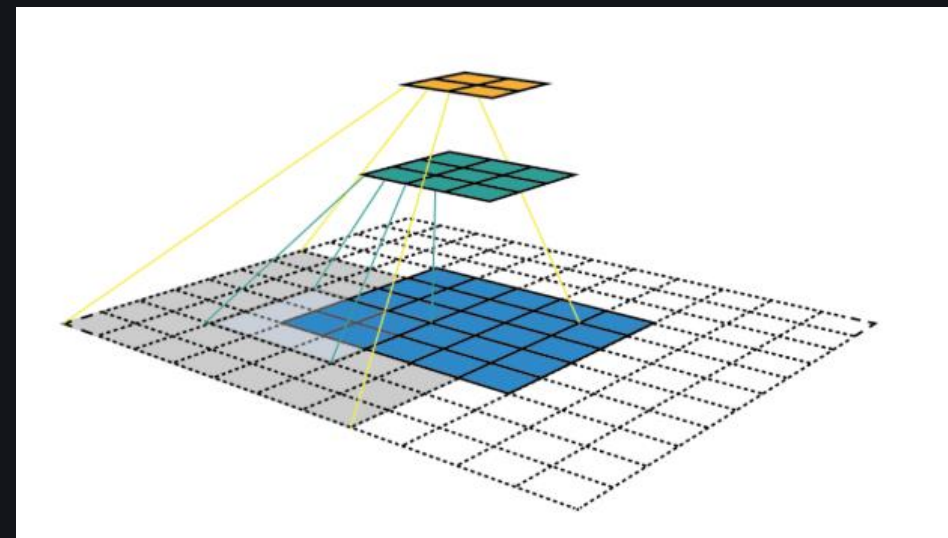
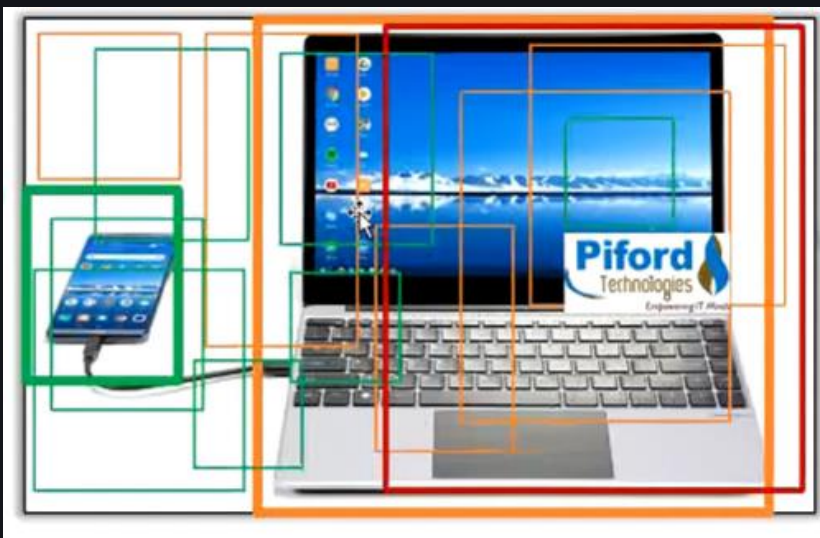
Input treningowy



Image classification



Object detection



Steps in Single Shot Detection

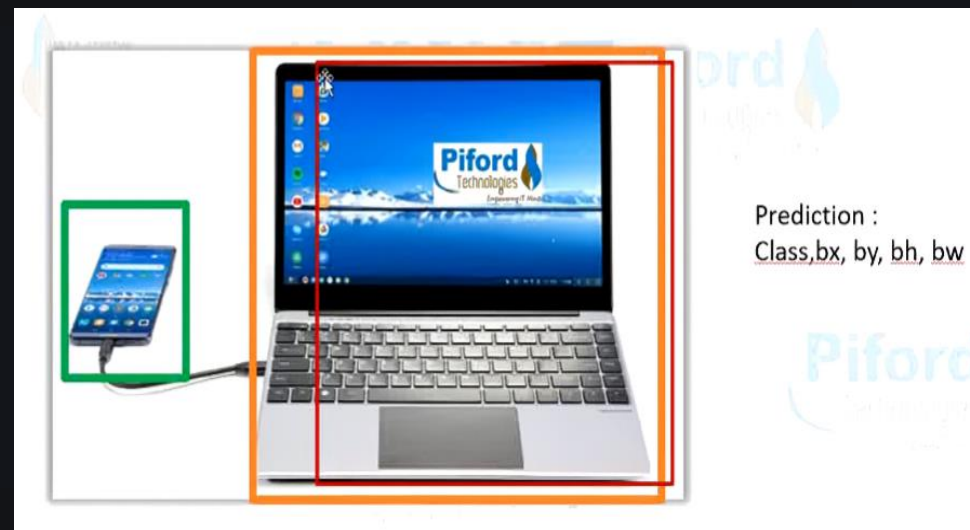
1. **Input Image:** The image is passed through the base network to extract feature maps.
2. **Feature Extraction:** The extra layers process these maps at multiple scales.
3. **Default Boxes Assignment:** Default boxes of various sizes and aspect ratios are assigned to each feature map cell.
4. **Prediction:** For each default box, class scores and bounding box offsets are predicted.
5. **Loss Calculation:** The loss is computed based on localization and confidence.
6. **NMS:** Redundant boxes are removed to produce the final set of detections.

Dla każdego z 8732 anchorów pytamy: czy ten anchor "odpowiada" za ten GT box?

- Anchor z $\text{IoU} > 0.5 \rightarrow$ **pozytywny** $\rightarrow$ "Ty masz wykryć ten obiekt"
- Anchor z $\text{IoU} < 0.5 \rightarrow$ **negatywny** $\rightarrow$ "Ty masz powiedzieć tło"

Non-Maximum Suppression (NMS)

To finalize the detection process, SSD applies Non-Maximum Suppression (NMS). This step eliminates redundant boxes with lower confidence scores, ensuring that only the most confident and relevant predictions are retained.



Key Advantages of SSD

1. **Speed:** One of the primary advantages of SSD is its speed. By eliminating the need for a region proposal network, SSD performs detection in a single shot, making it significantly faster than region-based algorithms like Faster R-CNN.
2. **Simplicity:** SSD's straightforward architecture simplifies the detection process. The single-pass approach reduces complexity and makes the network easier to train and implement.
3. **Accuracy:** SSD achieves competitive accuracy, especially for large objects, due to its multi-scale approach. By utilizing feature maps from multiple layers, SSD effectively detects objects of varying sizes.